

CPU Architecture & Register Organization

Q. Explain the internal organization of a CPU. Describe the basic components including Registers, ALU, and the Control Unit.

****Definition to Remember****

The **CPU** executes instructions (processes data).

1. Registers

High-speed, temporary storage locations situated directly on the CPU chip.

* **General Purpose Registers**

2. Arithmetic and Logic Unit (ALU)

The execution core where actual data processing happens.

* **Arithmetic Logic Unit (ALU)**

3. Control Unit (CU)

The manager of the CPU. It does not process data itself.

* **Fetch** memory

4. Internal CPU Organization

The Registers, ALU, and CU are connected via an Internal CPU Bus. This localized data pathway ensures high-speed data transfer between CPU components without having to use the slower external system bus.

****Must-Write Points (for 10 marks)****

1. CPU

****Quick Recall****

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